Introduction lists(从头是0，结尾是-1)

In [11]:
```
bicycles = ['trek', 'cannondale', 'redline', 'specialized']
print(bicycles)
```

```
['trek', 'cannondale', 'redline', 'specialized']
```

In [10]:
```
print(bicycles[:]) # 输出整个列表
print(bicycles[1:3]) # python是从0开始，左闭右开(不含3)
print(bicycles[1:]) # 从1到结尾
print(bicycles[:3]) # 从 0-3(不含3)
print(bicycles[-1]) # 从右到左，右边最后一个数(尾数从-1开始)
print(bicycles[-2].title())
```

```
['trek', 'cannondale', 'redline', 'specialized']
['cannondale', 'redline']
['cannondale', 'redline', 'specialized']
['trek', 'cannondale', 'redline']
specialized
Redline
```

Using Individual list Values(使用list某个值)

In [12]:
```
bicycles = ['trek', 'cannondale', 'redline', 'specialized']
message="My first bicycle was a " + bicycles[2].title() + "!"
print(message)
```

```
My first bicycle was a Redline!
```

Excercise

In [13]:
```
# 3-1
names=['wenda','anna','lisa']

print(names[0].title())
print(names[1].title())
print(names[2].title())
```

```
Wenda
Anna
Lisa
```

In [14]:
```
# 3-2
names=['wenda','anna','lisa']

message_0="Hello " +names[0].title() + ", you are my best friend!"
message_1="Hello " +names[1].title() + ", you are my best friend!"
message_2="Hello " +names[2].title() + ", you are my best friend!"

print(message_0)
print(message_1)
print(message_2)
```

```
Hello Wenda, you are my best friend!
Hello Anna, you are my best friend!
Hello Lisa, you are my best friend!
```

In [15]:
```python
# 3-3:
transportations=['honda','bmw','subway']

message_0=transportations[0].title() + " is a motorcycle!"
message_1=transportations[1].title() + " is a car!"
message_2=transportations[2].title() + " is a public transportation!"

print(message_0)
print(message_1)
print(message_2)
```

```
Honda is a motorcycle!
Bmw is a car!
Subway is a public transportation!
```

Modifying Elements in a List

In [16]:
```python
motorcycles = ['honda', 'yamaha', 'suzuki']
print(motorcycles)

motorcycles[0]='ducati'
print(motorcycles)
```

```
['honda', 'yamaha', 'suzuki']
['ducati', 'yamaha', 'suzuki']
```

append()--在list最后增加

In [17]:
```python
motorcycles = ['honda', 'yamaha', 'suzuki']
print(motorcycles)

motorcycles.append('ducati')
print(motorcycles)
```

```
['honda', 'yamaha', 'suzuki']
['honda', 'yamaha', 'suzuki', 'ducati']
```

In [18]:
```python
motorcycles =[]
motorcycles.append('yamaha')
motorcycles.append('ducati')
motorcycles.append('honda')
print(motorcycles)
```

```
['yamaha', 'ducati', 'honda']
```

insert(位置，‘元素’)--在list中插入元素

In [19]:
```python
motorcycles = ['honda', 'yamaha', 'suzuki']
motorcycles.insert(1,'ducati')
print(motorcycles)
```

```
['honda', 'ducati', 'yamaha', 'suzuki']
```

Removing Elements from a List

del --按索引删除

In [38]:
```
motorcycles = ['honda', 'yamaha', 'suzuki']
print(motorcycles)

del motorcycles[1]
print(motorcycles)
```

```
['honda', 'yamaha', 'suzuki']
['honda', 'suzuki']
```

pop()--从最后删除

In [39]:
```
motorcycles = ['honda', 'yamaha', 'suzuki']
print(motorcycles)

motorcycles.pop()
print(motorcycles)
```

```
['honda', 'yamaha', 'suzuki']
['honda', 'yamaha']
```

In [41]:
```
motorcycles = ['honda', 'yamaha', 'suzuki']
last_owned = motorcycles.pop()
print("The last motorcycle i owned was a " + last_owned.title() + ".")
```

```
The last motorcycle i owned was a Suzuki.
```

In [22]:
```
motorcycles = ['honda', 'yamaha', 'suzuki']
second_owned = motorcycles.pop(1)
print("The second motorcycle i owned was a " + second_owned.title() + ".")
```

```
The second motorcycle i owned was a Yamaha.
```

remove()--移除指定元素

In [46]:
```
motorcycles = ['honda', 'yamaha', 'suzuki', 'ducati']
print(motorcycles)

motorcycles.remove('suzuki')
print(motorcycles)
```

```
['honda', 'yamaha', 'suzuki', 'ducati']
['honda', 'yamaha', 'ducati']
```

In [24]:
```
motorcycles = ['honda', 'yamaha', 'suzuki', 'ducati']
print(motorcycles)

motorcycles.remove(motorcycles[1])
print(motorcycles)
```

```
['honda', 'yamaha', 'suzuki', 'ducati']
['honda', 'suzuki', 'ducati']
```

In [25]:
```python
motorcycles = ['honda', 'yamaha', 'suzuki', 'ducati']
print(motorcycles)

too_expensive='yamaha'
motorcycles.remove(too_expensive)
print(motorcycles)
print("\nA " + too_expensive.title() + " is too expensive for me.")
```

```
['honda', 'yamaha', 'suzuki', 'ducati']
['honda', 'suzuki', 'ducati']

A Yamaha is too expensive for me.
```

Excercise

In [26]:
```python
# 3-4
guests=['wenda','jack','anna','nina']
print(guests[0].title() + ", would you like to have a dinner with me ?")
print(guests[1].title() + ", would you like to have a dinner with me ?")
print(guests[2].title() + ", would you like to have a dinner with me ?")
print(guests[3].title() + ", would you like to have a dinner with me ?")
```

```
Wenda, would you like to have a dinner with me ?
Jack, would you like to have a dinner with me ?
Anna, would you like to have a dinner with me ?
Nina, would you like to have a dinner with me ?
```

In [28]:
```python
# 3-5
guests=['wenda','jack','anna','nina']
print(guests[0].title() + ", would you like to have a dinner with me ?")
print(guests[1].title() + ", would you like to have a dinner with me ?")
print(guests[2].title() + ", would you like to have a dinner with me ?")
print(guests[3].title() + ", would you like to have a dinner with me ?")

print("\n" + guests[3].title() + " doesn't have time to have a dinner with me.")

guests[3]='susan'
new_guests=guests
print("\nNew guests are as following:")
print(new_guests)

print("\nA second set of invitation messages:")
print(new_guests[0].title() + ", would you like to have a dinner with me ?")
print(new_guests[1].title() + ", would you like to have a dinner with me ?")
print(new_guests[2].title() + ", would you like to have a dinner with me ?")
print(new_guests[3].title() + ", would you like to have a dinner with me ?")
```

```
Wenda, would you like to have a dinner with me ?
Jack, would you like to have a dinner with me ?
Anna, would you like to have a dinner with me ?
Nina, would you like to have a dinner with me ?

Nina doesn't have time to have a dinner with me.

New guests are as following:
['wenda', 'jack', 'anna', 'susan']

A second set of invitation messages:
Wenda, would you like to have a dinner with me ?
Jack, would you like to have a dinner with me ?
Anna, would you like to have a dinner with me ?
Susan, would you like to have a dinner with me ?
```

In [37]:

```python
# 3-6
guests=['wenda','jack','anna','nina']
print(guests[0].title() + ", would you like to have a dinner with me ?")
print(guests[1].title() + ", would you like to have a dinner with me ?")
print(guests[2].title() + ", would you like to have a dinner with me ?")
print(guests[3].title() + ", would you like to have a dinner with me ?")

print("\n" + guests[3].title() + " doesn't have time to have a dinner with me.")

guests[3]='susan'
new_guests=guests
print("\nNew guests are as following:")
print(new_guests)

print("\nA second set of invitation messages:")
print(new_guests[0].title() + ", do you have time to have a dinner with me ?")
print(new_guests[1].title() + ", do you have time to have a dinner with me ?")
print(new_guests[2].title() + ", do you have time to have a dinner with me ?")
print(new_guests[3].title() + ", do you have time to have a dinner with me ?")

print("\nHi, my dear friend, I found a bigger dinner table!")

new_guests.insert(0,'jenny')
new_guests.insert(2,'donna')
new_guests.append('sofia')

print("\nUpdate new guests list:")
print(new_guests)

print("\nA Third set of invitation messages:")
print(new_guests[0].title() + ", let's have a dinner.")
print(new_guests[1].title() + ", let's have a dinner.")
print(new_guests[2].title() + ", let's have a dinner.")
print(new_guests[3].title() + ", let's have a dinner.")
print(new_guests[4].title() + ", let's have a dinner.")
print(new_guests[5].title() + ", let's have a dinner.")
print(new_guests[6].title() + ", let's have a dinner.")
```

```
Wenda, would you like to have a dinner with me ?
Jack, would you like to have a dinner with me ?
Anna, would you like to have a dinner with me ?
Nina, would you like to have a dinner with me ?

Nina doesn't have time to have a dinner with me.

New guests are as following:
['wenda', 'jack', 'anna', 'susan']

A second set of invitation messages:
Wenda, do you have time to have a dinner with me ?
Jack, do you have time to have a dinner with me ?
Anna, do you have time to have a dinner with me ?
Susan, do you have time to have a dinner with me ?

Hi, my dear friend, I found a bigger dinner table!

Update new guests list:
['jenny', 'wenda', 'donna', 'jack', 'anna', 'susan', 'sofia']

A Third set of invitation messages:
Jenny, let's have a dinner.
Wenda, let's have a dinner.
Donna, let's have a dinner.
Jack, let's have a dinner.
Anna, let's have a dinner.
Susan, let's have a dinner.
Sofia, let's have a dinner.
```

In [38]:
```
# 3-7
print("Sorry friends,I have space for only two guests.")
print(new_guests)
```

```
Sorry friends,I have space for only two guests.
['jenny', 'wenda', 'donna', 'jack', 'anna', 'susan', 'sofia']
```

In [39]:
```
print("Sorry " + new_guests.pop().title() + ", I am sorry can't invite you to di
```

```
Sorry Sofia, I am sorry can't invite you to dinner.
```

In [40]:
```
print("Sorry " + new_guests.pop().title() + ", I am sorry can't invite you to di
```

```
Sorry Susan, I am sorry can't invite you to dinner.
```

In [41]:
```
print("Sorry " + new_guests.pop().title() + ", I am sorry can't invite you to di
```

```
Sorry Anna, I am sorry can't invite you to dinner.
```

In [42]:
```
print("Sorry " + new_guests.pop().title() + ", I am sorry can't invite you to di
```

```
Sorry Jack, I am sorry can't invite you to dinner.
```

In [43]:
```
print("Sorry " + new_guests.pop().title() + ", I am sorry can't invite you to di
```

```
Sorry Donna, I am sorry can't invite you to dinner.
```

In [44]:
```
print(new_guests)
print(new_guests[0].title() + ", you are still invited.")
print(new_guests[1].title() + ", you are still invited.")
```

```
['jenny', 'wenda']
Jenny, you are still invited.
Wenda, you are still invited.
```

In [45]:
```python
del new_guests[:]
print(new_guests)
```

```
[]
```

sort()--按字母顺序输出

In [47]:
```python
# 正序(按字母顺序)
cars = ['bmw', 'audi', 'toyota', 'subaru']
cars.sort() # 永久
print(cars)
```

```
['audi', 'bmw', 'subaru', 'toyota']
```

In [48]:
```python
# 逆序(按字母顺序)
cars = ['bmw', 'audi', 'toyota', 'subaru']
cars.sort(reverse=True)
print(cars)
```

```
['toyota', 'subaru', 'bmw', 'audi']
```

sorted()--不改变原列表顺序

In [50]:
```python
cars = ['bmw', 'audi', 'toyota', 'subaru']
print("Here is the original list:")
print(cars)

print("\nHere is the sorted list:")
print(sorted(cars)) # sorted(cars)临时改变顺序，如果用cars.sort()则是永久改变顺序

print("\nHere is the original list again:")
print(cars)
```

```
Here is the original list:
['bmw', 'audi', 'toyota', 'subaru']

Here is the sorted list:
['audi', 'bmw', 'subaru', 'toyota']

Here is the original list again:
['bmw', 'audi', 'toyota', 'subaru']
```

In [55]:
```python
cars = ['bmw', 'audi', 'toyota', 'subaru']

# 按字母倒序输出
result = sorted(cars, reverse=True)
print(result)
```

```
['toyota', 'subaru', 'bmw', 'audi']
```

reverse()--逆序输出(不管字母顺序)

In [56]:
```
cars = ['bmw', 'audi', 'yamha', 'subaru']

# 原始顺序输出
print(cars)

# 逆序输出
cars.reverse()
print(cars)

# 再逆序一次就是原始顺序输出
cars.reverse()
print(cars)
```

```
['bmw', 'audi', 'yamha', 'subaru']
['subaru', 'yamha', 'audi', 'bmw']
['bmw', 'audi', 'yamha', 'subaru']
```

len()

In [57]:
```
cars = ['bmw', 'audi', 'toyota', 'subaru']
len(cars)
```

Out[57]: 4

Excercise

In [59]:
```
# 3-8
places=['kuala lumpur','new york','london','Beijing','shanghai']

# 原顺序输出
print("Original order lists:")
print(places)

# 按字母正序输出(不改变原列表)
print("\nSorted order lists:")
print(sorted(places))

# 原顺序输出
print("\nOriginal order lists again:")
print(places)

# 按字母逆序输出(不改变原列表)
print("\nReverse alphabetical order lists:")
print(sorted(places,reverse=True))

# 原顺序输出
print("\nOriginal order lists:")
print(places)

# 列表反转（颠倒原有顺序，不按字母）
print("\nUse reverse() to change the lists:")
places.reverse()
print(places)

# 再次反转，恢复原来顺序
```

```python
print("\nBack to its original order:")
places.reverse()
print(places)

# 按字母正序排列（永久修改原列表）
print("\nStored in alphabetical order:")
places.sort()
print(places)

# 按字母逆序排列（永久修改原列表）
print("\nStored in reverse alphabetical order:")
places.sort(reverse=True)
print(places)
```

```
Original order lists:
['kuala lumpur', 'new york', 'london', 'Beijing', 'shanghai']

Sorted order lists:
['Beijing', 'kuala lumpur', 'london', 'new york', 'shanghai']

Original order lists again:
['kuala lumpur', 'new york', 'london', 'Beijing', 'shanghai']

Reverse alphabetical order lists:
['shanghai', 'new york', 'london', 'kuala lumpur', 'Beijing']

Original order lists:
['kuala lumpur', 'new york', 'london', 'Beijing', 'shanghai']

Use reverse() to change the lists:
['shanghai', 'Beijing', 'london', 'new york', 'kuala lumpur']

Back to its original order:
['kuala lumpur', 'new york', 'london', 'Beijing', 'shanghai']

Stored in alphabetical order:
['Beijing', 'kuala lumpur', 'london', 'new york', 'shanghai']

Stored in reverse alphabetical order:
['shanghai', 'new york', 'london', 'kuala lumpur', 'Beijing']
```

In [60]:

```python
# 3-9
guests=['jenny', 'wenda', 'donna', 'jack', 'anna', 'susan', 'sofia']
print("I invite " + str(len(guests)) + " friends to dinner.")
```

```
I invite 7 friends to dinner.
```

In [61]:

```python
# 3-10 Every Function
# 创建一个列表（我用城市列表）
cities = ['tokyo', 'paris', 'london', 'sydney', 'dubai']

# 1. 打印原始列表
print("Original list:")
print(cities)

# 2. 访问列表元素（索引）
print("\nFirst city:", cities[0].title())

# 3. 修改元素
cities[1] = 'rome'
print("\nAfter changing city:", cities)
```

```python
# 4. 添加元素到末尾 append()
cities.append('berlin')
print("\nAfter append:", cities)

# 5. 插入元素到指定位置 insert()
cities.insert(2, 'madrid')
print("\nAfter insert:", cities)

# 6. 删除元素 del
del cities[0]
print("\nAfter del:", cities)

# 7. 弹出末尾元素 pop()
popped_city = cities.pop()
print("\nPopped city:", popped_city)
print("Now list:", cities)

# 8. 根据值删除 remove()
cities.remove('madrid')
print("\nAfter remove:", cities)

# 9. 临时排序 sorted()
print("\nSorted list:", sorted(cities))

# 10. 永久排序 sort()
cities.sort()
print("\nAfter sort():", cities)

# 11. 反向永久排序 sort(reverse=True)
cities.sort(reverse=True)
print("\nAfter sort(reverse=True):", cities)

# 12. 反转列表 reverse()
cities.reverse()
print("\nAfter reverse():", cities)

# 13. 获取列表长度 len()
print("\nTotal number of cities:", len(cities))
```

```
Original list:
['tokyo', 'paris', 'london', 'sydney', 'dubai']

First city: Tokyo

After changing city: ['tokyo', 'rome', 'london', 'sydney', 'dubai']

After append: ['tokyo', 'rome', 'london', 'sydney', 'dubai', 'berlin']

After insert: ['tokyo', 'rome', 'madrid', 'london', 'sydney', 'dubai', 'berlin']

After del: ['rome', 'madrid', 'london', 'sydney', 'dubai', 'berlin']

Popped city: berlin
Now list: ['rome', 'madrid', 'london', 'sydney', 'dubai']

After remove: ['rome', 'london', 'sydney', 'dubai']

Sorted list: ['dubai', 'london', 'rome', 'sydney']

After sort(): ['dubai', 'london', 'rome', 'sydney']

After sort(reverse=True): ['sydney', 'rome', 'london', 'dubai']

After reverse(): ['dubai', 'london', 'rome', 'sydney']

Total number of cities: 4
```

In [63]:

```python
# 3-11
cities = ['tokyo', 'paris', 'london', 'sydney', 'dubai']
print(cities[4])
```

```
dubai
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: